

Digital Micro-Intervention for Health Anxiety (Hypochondriasis)

A HAPA-based approach to disrupting hypochondria and
cyberchondria

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Abstract

This project presents STEADY, an innovative digital health application designed to disrupt the maladaptive cycle of cyberchondria through the use of Just-in-Time Adaptive Interventions (JITAI). The project is grounded in the Health Action Process Approach (HAPA) model, specifically targeting the volitional phase to bridge the "intention-behavior gap" that prevents individuals with health anxiety from refraining from compulsive online symptom checking. By implementing "if-then" coping planning, the application enables users to automate their responses to high-anxiety triggers, effectively reducing cognitive load during critical states of vulnerability. The system architecture, utilizing the React Native/Expo framework, ensures secure data processing and continuous learning based on both active and passive user inputs. This project demonstrates that focused digital micro-interventions, oriented toward enhancing self-efficacy, can effectively support self-regulation and serve as a valuable tool in modern anxiety management.

Introduction

Problem Definition

The rapid and explosive growth of the Internet has changed the way individuals obtain information regarding health and prevention (Vismara et al., 2020). While the benefits of accessibility are evident, it also presents significant challenges for individuals prone to health anxiety (Vismara et al., 2017). Health anxiety (HA) is defined as a multidimensional phenomenon that arises when bodily sensations or physiological changes are interpreted as signs of a serious medical condition (Lebel et al., 2020; Vismara et al., 2020). Historically, these conditions were clinically classified as hypochondriasis (Escobar et al., 1998; Vismara et al., 2020). The International Classification of Diseases (ICD-10) categorized hypochondriasis as a mental and behavioral disorder, whereas the DSM-IV-TR classification defined it as a somatoform disorder (Escobar et al., 1998; Vismara et al., 2020). Studies conducted in primary care settings have shown that this condition is a significant problem, affecting approximately 3.4% of all patients.

A significant shift in psychiatric classification occurred in 2013 with the publication of the DSM-5; this document officially replaced the diagnosis of hypochondriasis with two new, distinct categories that better capture the diversity of patient experiences (Lebel et al., 2020). It is estimated that, under the current system, 75% of individuals previously diagnosed with hypochondriasis now meet the criteria for somatic symptom disorder, which is characterized by distressing physical symptoms accompanied by excessive thoughts and behaviors (Lebel et al., 2020). The remaining 25% of patients fall into the category of illness anxiety disorder; this condition manifests primarily as persistent worry about the possibility of having or

developing a serious illness, in the absence of significant somatic symptoms (Lebel et al., 2020). This evolution reflects a shift toward viewing health anxiety along a continuum—ranging from mild and transient to severe and chronic—depending on the impact these concerns have on an individual's quality of life and daily functioning (Lebel et al., 2020).

In this digital context, when health concerns lead to compulsive and excessive online information-seeking, the behavior is termed "cyberchondria" (McMullan et al., 2019; Vismara et al., 2020). Cyberchondria functions as a form of "digital safety-seeking behavior" in which an individual attempts to neutralize anxiety by repeatedly checking for information—a process that paradoxically results in heightened stress and an exacerbation of the initial health concerns (McMullan et al., 2019; Vismara et al., 2020).

Beyond individual distress, cyberchondria represents a significant public health burden with substantial economic implications (Vismara et al., 2020). Research indicates that individuals with high health anxiety represent a significant economic burden due to increased healthcare utilization and frequent requests for medical consultations (Vismara et al., 2020). In the UK alone, direct and indirect costs associated with health anxiety have been estimated at £56 million per year (Vismara et al., 2020). Furthermore, excessive online searches can lead to unnecessary medical tests and a deterioration in the doctor-patient relationship, as patients may prioritize algorithm-driven results over professional medical advice (Vismara et al., 2020).

The Cyberchondria Loop

The mechanism of cyberchondria is conceptualized as a self-perpetuating, maladaptive feedback loop in which digital behaviors—intended to alleviate uncertainty—paradoxically intensify health anxiety (Martino et al., 2026; Vismara et al., 2020). This cycle typically begins with a trigger—such as an ambiguous bodily sensation or a health concern—that gives rise to a compulsive need for reassurance through online information seeking (Singh et al., 2016; Vismara et al., 2020). In contrast to adaptive health information seeking, this “digital safety-seeking behavior” stems from a dispositional intolerance of uncertainty and an inability to achieve lasting peace of mind (Martino et al., 2026; McMullan et al., 2019).

A core cognitive driver of this cycle is the dispositional intolerance of uncertainty (IU), a bias where individuals find medical ambiguity unacceptable and threatening regardless of the actual probability of illness (Martino et al., 2026; McMullan et al., 2019). Individuals with high health anxiety are driven by a compulsive need for “perfect” or exhaustive explanations for their symptoms (Martino et al., 2026). However, since unregulated online health information is often inaccurate, misleading, or incomplete, it fails to provide lasting reassurance (McMullan et al., 2019). Instead of resolving doubt, the informational abundance fuels further uncertainty, compelling the user to initiate even more detailed searches to resolve the resulting epistemic instability (Martino et al., 2026; McMullan et al., 2019).

The process of seeking information often involves “query escalation,” a stage where searching for details about common or benign symptoms (e.g., a headache) frequently leads to encountering references to sensational or rare yet life-threatening conditions (e.g., a brain tumor), thereby significantly heightening the sense of threat

(McMullan et al., 2019; Singh et al., 2016). This process is further fueled by the confirmation bias, which causes individuals to selectively focus on information that validates their catastrophic fears while ignoring benign medical explanations (Martino et al., 2026; McMullan et al., 2019).

A key element sustaining this cycle is the paradox of temporary relief; cognitive-behavioral models suggest that checking information online can lead to a short-term reduction in anxiety (Singh et al., 2016; Vismara et al., 2020).

This fleeting relief operates through negative reinforcement, entrenching the habit and making it harder to resist the urge to seek information in the future (Vismara et al., 2020).

However, this sense of relief is quickly undermined by the ambiguity and contradictory nature of unregulated online content, giving rise to new doubts and a phenomenon researchers term "information entropy" (Martino et al., 2026; McMullan et al., 2019).

Ultimately, this loop culminates in heightened anxiety: individuals with high levels of illness anxiety report feeling significantly worse after checking symptoms online, whereas those with low anxiety levels may experience relief (Doherty-Torstrick et al., 2016).

This repetitive process reinforces dysfunctional beliefs about illness and contributes to a significant decline in functioning, as the individual becomes trapped in an increasingly stressful cycle of constant symptom checking and a lack of reassurance (Doherty-Torstrick et al., 2016; Vismara et al., 2020). To better understand the self-perpetuating dynamics of this process, Figure 1 illustrates the

course of the cyberchondria cycle, highlighting how safety-seeking behaviors lead to information entropy rather than a resolution of the problem.

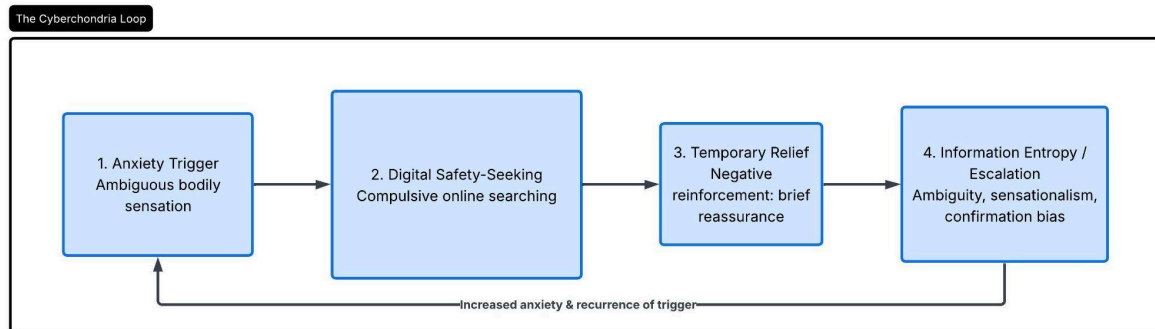


Figure 1. The Cyberchondria Loop. A schematic representation of the self-perpetuating cycle of health anxiety driven by digital safety-seeking behaviors.

Objectives of the project

The primary objective of this project is to develop a digital tool that disrupts the maladaptive cycle of cyberchondria by leveraging the unique capabilities of Just-in-Time Adaptive Interventions (JITAI) and micro-interventions. Digital mental health solutions hold particular promise for addressing the global mental health crisis, as they overcome traditional barriers to care—such as high costs, geographical isolation, and the stigma associated with seeking professional help (Kornacka et al., 2025; Maciejewski & Smoktunowicz, 2023; Van Ameringen et al., 2017). Given that cyberchondria is rooted in digital behaviors, a mobile intervention allows researchers to "meet participants where they already spend their time," providing support directly within the user's natural environment (Maciejewski & Smoktunowicz, 2023).

A key rationale for an approach based on low-effort micro-interventions is the discrepancy between the intervention's design assumptions and actual user

behavior. Research indicates that most users of self-guided mental health apps do not engage with them extensively, and user retention rates often drop significantly after the first two weeks (Baumel et al., 2020).

The rationale for a low-effort approach is also grounded in the need to minimize cognitive load during states of acute distress. In moments of high anxiety, individuals often experience impaired executive functioning, particularly in areas of inhibitory control and decision-making, which makes engaging with complex, traditional therapeutic tasks difficult (Martino et al., 2026). By providing highly focused, brief units of support, micro-interventions lower the "bar for entry" and reduce the energy required for purposeful engagement (Baumel et al., 2020). This resource efficiency is designed to intercept compulsive impulses without overwhelming the user's limited cognitive resources, thereby reducing mental fatigue and preventing habituation to the intervention (Baumel et al., 2020; Nahum-Shani & Murphy, 2026).

Moreover, the JITAI model enables interventions to be delivered during specific "states of vulnerability"—that is, precisely when the user feels a compulsive urge to search for information about symptoms online (Nahum-Shani & Murphy, 2026; Naughton, 2016). Unlike traditional therapy, which takes place after the fact, digital tools can provide real-time feedback and support self-monitoring, making it possible to halt information-seeking behavior before the cycle of negative reinforcement is completed (Martino et al., 2026; Vismara et al., 2020).

Furthermore, the project aligns with the principles of the HAPA (Health Action Process Approach) model by focusing on building self-efficacy—a key factor determining whether an individual will undertake and sustain the effort to refrain

from behaviors driven by health anxiety (Fulton et al., 2019). As shown in Figure 2, this mechanism involves providing support precisely when the user reaches a "Vulnerability State," effectively transforming the process from a cycle of compulsive checking into an adaptive, self-regulating response based on the HAPA model.

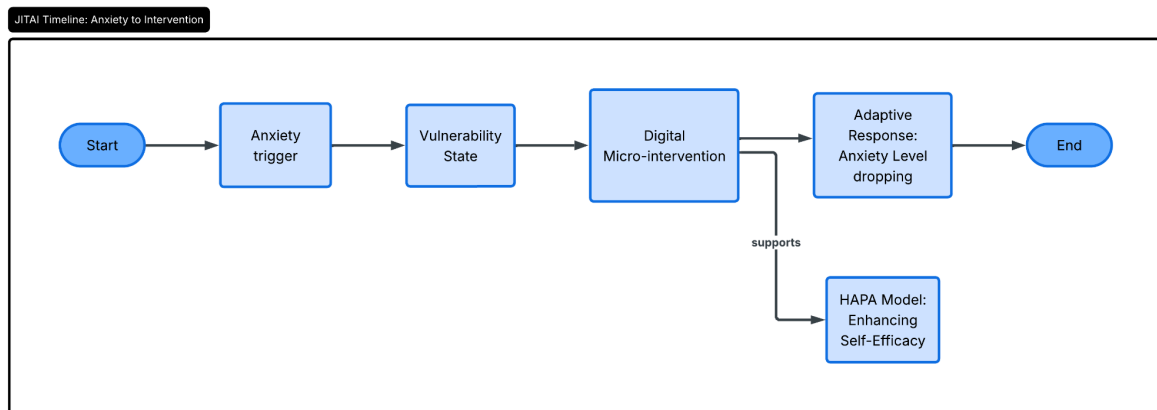


Figure 2. JITAI timeline: from anxiety to intervention. The diagram illustrates how a just-in-time adaptive intervention (JITAI) curbs the escalation of anxiety by providing support that fosters self-efficacy, in accordance with the HAPA model.

Theoretical framework (The “Why”)

The Hapa Model

The HAPA model (Health Action Process Approach) is a socio-cognitive theoretical framework describing the psychological processes leading to health behavior change (Schwarzer, 2008, 2016). Unlike traditional models that view behavior change as a linear continuum, HAPA distinguishes between two distinct phases: the motivational phase, in which the individual forms an intention to act, and the volitional phase, in which that intention is translated into actual behavior (Schwarzer, 2016; Sniehotta et al., 2005). The sequential nature of these processes

is illustrated in **Figure 3**, which shows the transition from the motivational phase to the volitional phase, marked by the key “intention-behavior gap.”

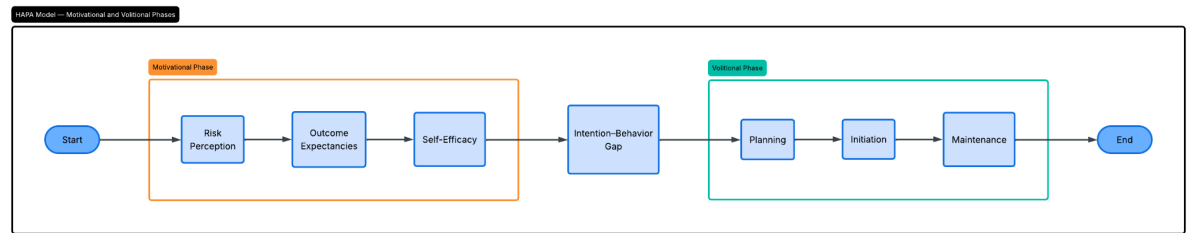


Figure 3. HAPA model – motivational and volitional phases. A flowchart illustrating the psychological process of transitioning from the initial formation of an intention to the successful maintenance of health behaviors, in which the "intention-behavior gap" serves as a key turning point.

The volitional phase (occurring after the intention has been formed) is central to the STEADY intervention, as it focuses on individuals who are already motivated to change but face difficulties in implementing that change (Schwarzer, 2016). This phase relies on three main self-regulatory constructs:

- **Volitional self-efficacy:** This is subdivided into maintenance self-efficacy (the belief in one's ability to overcome obstacles while pursuing the intended action) and recovery self-efficacy (the belief in one's ability to resume pursuing the goal after a setback, such as an episode of compulsive symptom-related information seeking) (Schwarzer, 2008, 2016; Sniehotta et al., 2005).
- **Planning:** This encompasses action planning (determining when, where, and how to act) and coping planning (anticipating specific high-risk situations that

trigger unwanted behavior and devising alternative ways of responding) (Hagger & Luszczynska, 2014; Schwarzer, 2016).

- **Action control (self-monitoring):** Research indicates that within the HAPA model, action control is often the most immediate predictor of behavior (Sniehotta et al., 2005). It involves the continuous comparison of current behavior against an adopted personal standard (e.g., "I intend not to check for symptoms"), thereby enabling real-time adjustments (Schwarzer, 2016; Sniehotta et al., 2005).

The Intention-Behavior Gap in Health Anxiety

A significant issue for individuals with health anxiety (HA) is the so-called intention-behavior gap—a state in which individuals have a strong desire to stop a specific behavior (e.g., excessive symptom checking) but fail to act on that intention when faced with a triggering stimulus (Hagger & Luszczynska, 2014). Such individuals are referred to as "inclined abstainers"—they possess the motivation but lack the self-regulation tools needed to resist impulses (Hagger & Luszczynska, 2014; Sniehotta et al., 2005).

In the case of cyberchondria, this gap is exacerbated by three reinforcing factors:

1. **Negative reinforcement:** Compulsive symptom checking often yields a short-term reduction in anxiety, serving as a powerful reward that reinforces the checking habit (Martino et al., 2026; Vismara et al., 2020). This immediate relief often overrides the long-term intention to remain calm (Vismara et al., 2020).

2. **Impaired inhibitory control:** States of intense anxiety are associated with a temporary decline in executive functions, particularly regarding the ability to inhibit impulsive actions (Martino et al., 2026). When a user experiences information entropy—a state of confusion caused by conflicting medical data—their cognitive resources are too depleted to rely solely on willpower (Martino et al., 2026).
3. **The “horse-race” metaphor:** An automatic, habitual impulse to seek reassurance “wins the race” against the slower, conscious intention to refrain from checking, especially when a concrete action plan is lacking (Hagger & Luszczynska, 2014).

Coping planning

To bridge the intention-behavior gap, the STEADY app implements coping planning based on so-called implementation intentions (Gollwitzer, 1999). These plans utilize a highly precise “if–then” logic: “If situation X (trigger) occurs, then I will take action Y (coping strategy)” (Hagger & Luszczynska, 2014; Schwarzer, 2016). This logical structure aims to simplify the decision-making process under stress, as illustrated in Figure 4. By linking a specific trigger to an adaptive response, the STEADY application helps users cope more effectively with stress vulnerability.

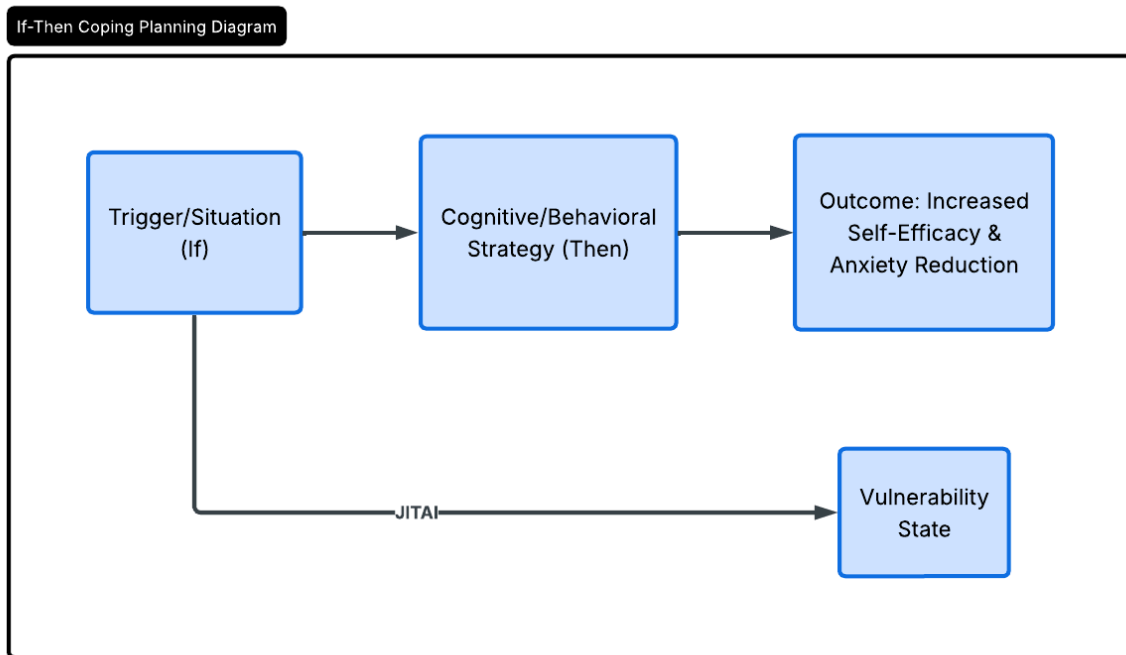


Figure 4. The 'If–Then' Coping Planning Mechanism. This diagram illustrates how the 'if–then' logic utilizes JITA! principles to intercept a trigger (vulnerability state) and pivot to a cognitive or behavioral strategy, thereby promoting self-efficacy and anxiety reduction .

This approach is theoretically grounded in several mechanisms:

- **Response automatization:** "If–then" plans shift control over behavior from the realm of conscious reflection to environmental cues (Hagger & Luszczynska, 2014). Through repeated practice of a specific response (e.g., "If I feel muscle tremors, then I will start a breathing exercise in the app"), the behavior becomes automatic and requires little effort—a crucial factor during states of intense anxiety when cognitive load is high (Baumel et al., 2020; Hagger & Luszczynska, 2014).

- **Building self-efficacy for behavior maintenance:** Successfully executing a pre-planned "if-then" strategy reinforces the user's belief in their ability to cope with triggers (Schwarzer, 2008, 2016). This fosters a sense of mastery (the experience of success), which is the strongest source of self-efficacy (Schwarzer, 2016).
- **Breaking the feedback loop:** Just-in-Time Adaptive Interventions (JITAI) utilize these decision rules to provide support precisely when the user is most vulnerable to difficulties (Nahum-Shani & Murphy, 2026). By intercepting the impulse before an information-seeking attempt occurs, the intervention prevents the completion of the negative reinforcement loop characteristic of cyberchondria, thereby weakening the habit (Martino et al., 2026; Vismara et al., 2020).

User Research & Requirements

User Personas: Evidence-Based Profiles

To ensure the STEADY app addresses real-world needs, three primary personas have been developed based on clinical profiles and behavioral patterns identified in research (Seiferth et al., 2023).

- **Persona 1: "The Inclined Abstainer" (The Regretful Searcher)**
 - **Theoretical Basis:** This persona is grounded in the intention-behavior gap described in the Health Action Process Approach (HAPA) model (Hagger & Luszczynska, 2014; Schwarzer, 2016).
 - **Profile:** These users possess a strong intention to stop compulsive symptom-checking but lack the volitional tools and self-regulatory

skills needed to resist impulses during high-anxiety states (Hagger & Luszczynska, 2014; Schwarzer, 2008). They often experience "post-session regret," where searching leads to significantly higher distress and anxiety than was present before the search was initiated (Doherty-Torstrick et al., 2016; Vismara et al., 2020).

- **Key Need:** Real-time interruption and action control strategies to bridge the gap between their long-term health goal and their immediate compulsive search actions (Schwarzer, 2016; Sniehotta et al., 2005).

- **Persona 2: "The Uncertainty Seeker" (The Digital Monitor)**

- **Theoretical Basis:** This persona is driven by dispositional intolerance of uncertainty (IU) and its associated cognitive biases (Martino et al., 2026; McMullan et al., 2019).
- **Profile:** Characterized by a compulsive need for "perfect," exhaustive, and coherent explanations for ambiguous bodily sensations (Martino et al., 2026). They engage in repetitive searches because unregulated online health information often fails to provide lasting reassurance, instead fueling further doubt (Martino et al., 2026; McMullan et al., 2019).
- **Key Need:** Functionalities to manage informational entropy and the cognitive instability generated by conflicting online data (Martino et al., 2026).

- **Persona 3: "The Escalator" (The High-Vulnerability User)**

- **Theoretical Basis:** This profile is based on the "query escalation" phenomenon observed in large-scale studies of medical Web searches (White & Horvitz, 2009).
- **Profile:** These individuals typically start by searching for benign symptoms (e.g., a headache) and rapidly migrate to sensational or rare life-threatening conditions (e.g., a brain tumor) (Singh et al., 2016; White & Horvitz, 2009). This behavior represents a form of digital safety-seeking that reinforces dysfunctional beliefs and results in significant functional impairment (McMullan et al., 2019; Vismara et al., 2020).
- **Key Need:** Immediate "pivoting" strategies delivered at the moment of escalation to interrupt the self-perpetuating feedback loop (Martino et al., 2026; White & Horvitz, 2009).

User Stories: Mapping Needs to Functionalities

Functionalities are defined through the lens of Behavior Change Techniques (BCTs) to enhance user engagement and volitional self-efficacy (Baumel et al., 2020; Maciejewski & Smoktunowicz, 2023). The mapping of user needs to specific app functionalities is presented in **Figure 5** below

Persona	Need	Solution/BCT
The Inclined Abstainer	Automate response to impulse	Crisis Plan
The Uncertainty	Visualize lack of reassurance	Reflective Journal
The Escalator	Interrupt escalation	Adaptive Notifications

Figure 5. User Stories and Functionalities. A matrix mapping the specific needs of each persona to evidence-based Behavior Change Techniques (BCTs) implemented within the STEADY app

1. **As a user (Inclined Abstainer)**, I want to trigger a pre-defined "Crisis Plan" when I feel the urge to check, so that I can automate my response and reduce the cognitive load required to resist the impulse (Baumel et al., 2020; Hagger & Luszczynska, 2014).
2. **As a user (Uncertainty Seeker)**, I want to log my triggers and the outcome of my searches in a Reflective Journal, so that I can visualize the lack of lasting reassurance and monitor my behavior against personal standards (Martino et al., 2026; Sniehotta et al., 2005).
3. **As a user (The Escalator)**, I want to receive Adaptive Notifications during my "vulnerability states," so that I can be reminded of my coping strategies at the exact moment of an impulse (Nahum-Shani & Murphy, 2026).
4. **As a user**, I want to visualize my progress in resisting search urges, so that I can build my maintenance self-efficacy and belief in my ability to manage health anxiety (Fulton et al., 2019; Schwarzer, 2016).

Key "Trigger Points": Identification of Intervention Moments

Using the Just-in-Time Adaptive Intervention (JITAI) framework, the app identifies critical moments where intervention is most effective (Nahum-Shani & Murphy, 2026).

- **The Sensation Trigger:** The moment an individual notices an ambiguous bodily sensation which activates dysfunctional health beliefs and initiates the loop (Heathcote et al., 2022; McMullan et al., 2019).
- **The "Google" Impulse:** The specific point where the user feels a compulsive drive to seek reassurance through online searching, representing a primary state of vulnerability (Nahum-Shani & Murphy, 2026; Vismara et al., 2020).
- **The Escalation Point:** When a search session shifts focus from common symptoms to rare or serious illnesses, often triggered by escalatory terminology in search results (White & Horvitz, 2009).
- **The Informational Overload:** When the user encounters conflicting or overwhelming medical data, leading to informational entropy and acute cognitive stress (Martino et al., 2026).

Figure 6 visualizes the journey of a user experiencing these trigger points and the corresponding interventions provided by the STEADY app.

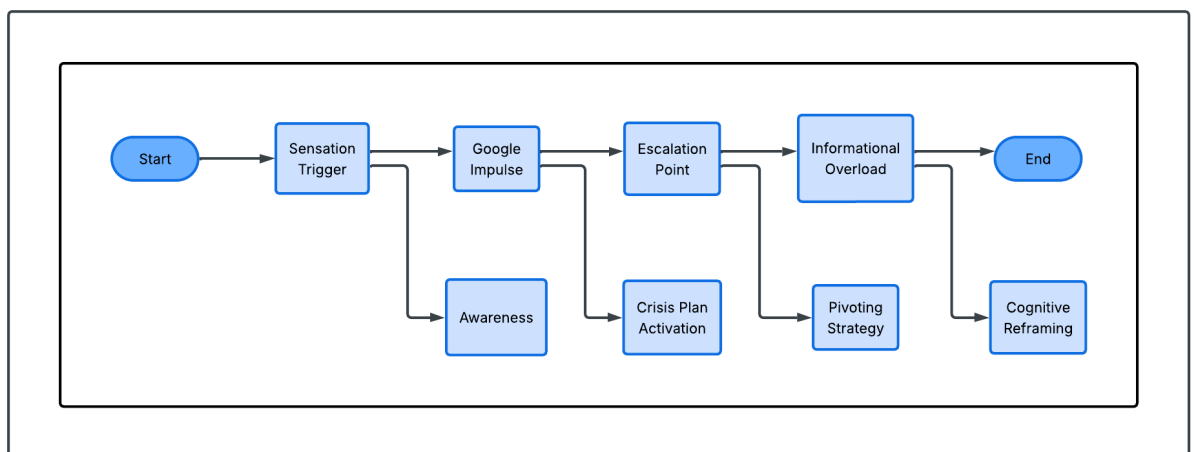


Figure 6. JITA! Trigger Points Journey. A visualization of critical intervention moments during a health anxiety episode, mapping user triggers to adaptive strategies provided by the system

Design & Prototyping (The "How")

User Flow: From Trigger to Coping Strategy

The user flow of the STEADY app is designed to map onto the cognitive-behavioral dynamics of health anxiety and the specific requirements of Just-in-Time Adaptive Interventions (JITAI) (Nahum-Shani & Murphy, 2026). The process begins with an anxiety trigger, typically an ambiguous bodily sensation that activates dysfunctional health beliefs (Heathcote et al., 2022; McMullan et al., 2019). This trigger leads the user into a state of vulnerability, characterized by a compulsive impulse to seek reassurance through online searches (Nahum-Shani & Murphy, 2026; Vismara et al., 2020). The STEADY app intercepts this cycle by delivering a digital micro-intervention at the exact moment the impulse occurs, providing real-time feedback (Baumel et al., 2020; Nahum-Shani & Murphy, 2026). Finally, the user is guided through an adaptive response, where anxiety levels drop as they pivot from a compulsive search to a pre-defined cognitive or behavioral coping strategy (Martino et al., 2026; Nahum-Shani & Murphy, 2026).

Core Functionalities: Justification of Features

The app's features are directly grounded in Behavior Change Techniques (BCTs) identified as effective in digital mental health interventions (Baumel et al., 2020; Maciejewski & Smoktunowicz, 2023).

- **Crisis Mode (IF-THEN Planning):** This feature implements implementation intentions, using a specific "if-then" logic to forge a mental link between a trigger and a healthy response (Hagger & Luszczynska, 2014). By specifying "If situation X occurs, then I will perform response Y," the app facilitates response automatization, allowing the healthy behavior to "win the race"

against habitual checking impulses (Hagger & Luszczynska, 2014; Schwarzer, 2016). This mechanism is crucial during high-anxiety states when the user's cognitive resources are too depleted for complex decision-making (Martino et al., 2026; Nahum-Shani & Murphy, 2026).

- **Reflective Journaling:** The journaling feature serves as a tool for action control (self-monitoring), which is the most proximal predictor of behavior change in the HAPA model (Sniehotta et al., 2005). Users log their triggers and the outcomes of their search urges, allowing them to visualize the lack of lasting reassurance provided by digital checking (Martino et al., 2026; Singh et al., 2016). This process builds maintenance self-efficacy by providing evidence of successful resistance to impulses, known as mastery experiences (Fulton et al., 2019; Schwarzer, 2016).
- **Adaptive Notifications:** To ensure ongoing engagement, notifications are designed to be low-effort, providing brief units of support that do not overwhelm the user (Baumel et al., 2020). These notifications are delivered based on JITAI principles, ensuring that support is provided during specific states of vulnerability or opportunity (Nahum-Shani & Murphy, 2026; Naughton, 2016). Research indicates that interactive notifications reinforcing HAPA constructs are more effective at supporting sustained commitment than generic, non-interactive reminders (Adorni et al., 2025).

UI/UX Principles: Support for Autonomy and Competence

The interface of the STEADY app is guided by Self-Determination Theory (SDT) and the need to reduce informational entropy (Adorni et al., 2025; Martino et al., 2026). The design prioritizes user autonomy by allowing individuals to self-define

their "if-then" plans and choose their preferred coping strategies (Adorni et al., 2025; Hagger & Luszczynska, 2014). To build a sense of competence, the app utilizes visual feedback and rewards, such as medals or progress charts, to reinforce the successful performance of healthy behaviors (Adorni et al., 2025; Alqahtani & Orji, 2019). Furthermore, the UI is optimized for low cognitive load, using simple navigation and clear instructions to accommodate individuals whose executive functions may be temporarily impaired by anxiety (Baumel et al., 2020; Martino et al., 2026).

Technical Implementation

Tech Stack: Justification for Chosen Tools

The choice of technology stack for the STEADY application is driven by the aim to maximize accessibility and minimize traditional barriers to mental healthcare, such as high costs or geographic exclusion. React Native and Expo were selected as the primary development environment, ensuring cross-platform availability (iOS and Android)—a crucial factor given that nearly two-thirds of the population currently use smartphones for health-related purposes. This decision aligns with the goal of creating an affordable, scalable solution that is easy to maintain and update, incorporating the latest evidence-based practices. Furthermore, the use of push notification systems within these technologies is essential for implementing Just-in-Time Adaptive Interventions (JITAI); this allows the application to reach users in their natural digital environment without requiring significant effort on their part. To ensure the system's credibility and reliability, its architecture must enable the

secure processing of sensitive health data, in compliance with emerging ethical and legal standards in the field of digital therapeutics.

System Architecture: High-Level Data Flow

The high-level system architecture of the STEADY application is designed to support the self-regulatory loops described in the HAPA model and to meet JITAI requirements for real-time operation. The data flow begins with user input, encompassing both active logging in the Reflection Journal and passive triggers activated when the user enters "Crisis Mode." This data is processed by the Intervention Planner, which applies predefined decision rules to determine the user's current "vulnerability state" or "capability state." If intervention criteria are met, the system initiates an adaptive notification or a specific "if-then" coping plan, providing immediate support to intercept compulsive behaviors. Finally, the results of these interactions are stored in a secure Intervention Documentation module, which feeds information back into the system to build a record of the user's skill mastery and track long-term progress in resisting the urge to check. This continuous data-driven loop is visualized in **Figure 7**.

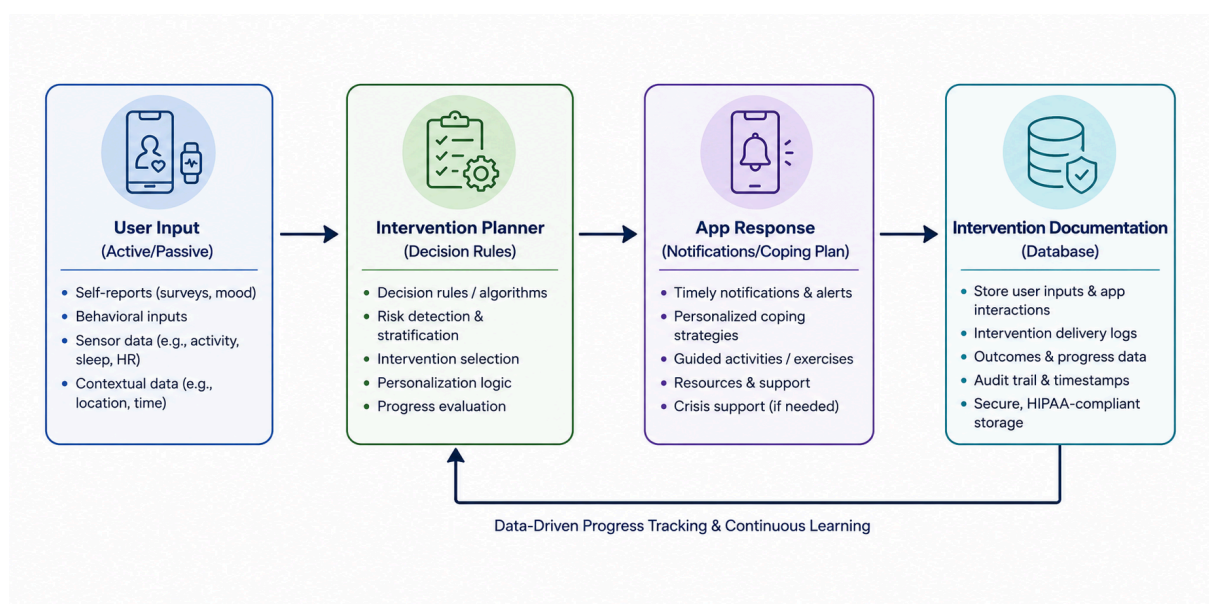


Figure 7. System Process Flow. A schematic representation of the STEADY application's data flow, illustrating the cycle from user input acquisition to the delivery of personalized interventions and long-term progress tracking

Discussion & Limitations

Ethical Considerations: Support vs. Replacement

The STEADY app is conceptualized as a supportive tool rather than a replacement for professional clinical intervention (Maciejewski & Smoktunowicz, 2023; Orzeł & Strzałek, 2025). While digital micro-interventions provide immediate assistance, research emphasizes that automated systems lack the human empathy and clinical intuition necessary for complex psychological processing (Minerva & Giubilini, 2023; Sweeney et al., 2021). Ethical implementation requires absolute transparency regarding the app's limitations, ensuring users understand that the software does not provide a medical diagnosis (Giacobello, 2025; Sweeney et al., 2021). Furthermore, the collection of sensitive health data through reflective journals necessitates strict adherence to data protection regulations (e.g., GDPR), as security breaches in mental health apps can lead to significant user harm (Sweeney et al., 2021; Zuryski et al., 2021). There is also a risk of dehumanization in therapy if users become overly dependent on algorithm-driven reassurance, which may paradoxically weaken the therapeutic alliance with human professionals (Holohan & Fiske, 2021; Martino et al., 2026).

Limitations and Future Potential

Several limitations of the current STEADY prototype must be acknowledged. First, like many mHealth interventions, the app relies heavily on self-report data,

which is susceptible to recall bias and the user's current emotional state (Doherty-Torstrick et al., 2016; McMullan et al., 2019). Second, the efficacy of the "Crisis Mode" in actual clinical populations remains to be verified, as existing research often utilizes convenience samples of university students rather than patients with diagnosed Illness Anxiety Disorder (Anagnostopoulos & Botse, 2016; Vismara et al., 2020). Additionally, while Just-in-Time Adaptive Interventions (JITAI) are promising, there is a risk of user habituation or "notification fatigue," where frequent prompts lead to decreased engagement over time (Nahum-Shani & Murphy, 2026). Future development should focus on conducting Randomized Controlled Trials (RCTs) to empirically validate the impact of "if-then" planning on cyberchondria severity (Kornacka et al., 2025; Starcevic, 2017). There is also potential to integrate wearable sensors to detect physiological indicators of anxiety, allowing for more objective triggering of interventions without increasing the user's manual reporting burden (Heathcote et al., 2022; Nahum-Shani & Murphy, 2026).

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